

KattidaCore — Application Workflow Reference

Mystics Civil construction operations cockpit.

This document describes every module in KattidaCore, the user roles that operate it, and the end-to-end workflows that move work from creation to completion. It is intended for project managers, administrators, and new team members who need a single, complete picture of how the platform behaves.

1. Platform Overview

KattidaCore is a multi-tenant web application for managing construction projects end-to-end. The platform is organised around three concepts:

Concept	Meaning
Organisation	A tenant — typically a contracting firm. All data is scoped to an organisation.
Project	A construction project belonging to one organisation.
User	A person with a login. Users belong to an organisation and hold one built-in role plus an optional custom role.

A user sees only the projects, modules and capabilities granted to them by their organisation, their role(s), and the project access list.

2. Identity, Roles & Access Control

2.1 Built-in roles (one per user)

The built-in role drives the major admin gates (who can manage users, who can approve, who bypasses per-project access checks).

Role	Purpose	Bypasses per-project access?
super_admin	Platform operator. Sees and manages every organisation.	Yes (across all orgs)
admin	Organisation administrator. Manages users, projects, roles in their org.	Yes (within org)
owner	Site / project owner. Strategic oversight, approvals.	Yes (within org)
pm	Project Manager. Day-to-day project execution.	No
site_engineer	Site execution, daily reports, attendance.	No
qs	Quantity Surveyor. BOQ, estimation, bill verification.	No
finance	Bills, payments, ledger, invoices.	No
contractor	External contractor. Submits DPRs, views own scope.	No
qc	Quality Control. Inspections, NCRs.	No
store	Storekeeper. Stock, GRN, issues.	No
hr	Workforce, payroll.	No
viewer	Read-only stakeholder.	No

2.2 Custom roles (optional, additive)

Administrators may define **org-scoped custom roles** (e.g. "Junior PM", "Site Auditor") with a hand-picked set of capabilities. A user's effective capability set is:

```
effective capabilities = preset(built-in role) ∪ custom role permissions
```

Custom roles **add** capabilities — they never remove them. Admins and super-admins implicitly hold every capability.

2.3 Capability catalog

The capability is the smallest enforceable permission. There are currently **13 capabilities**, grouped by domain:

Group	Capability	What it grants
Projects	<code>project:create</code>	Create new projects (filed for approval if not auto-approve role)
Projects	<code>project:transition</code>	Move a project through on-track / at-risk / delayed / on-hold
Projects	<code>project:complete</code>	Mark a project as completed
Approvals	<code>project:approve</code>	Approve or reject pending-approval projects
Approvals	<code>vo:approve</code>	Approve or reject variation orders
Approvals	<code>dpr:approve</code>	Approve or reject daily progress reports
DPRs	<code>dpr:create</code>	Draft and submit DPRs
DPRs	<code>dpr:edit</code>	Modify DPRs still in draft
Financial	<code>financial:view</code>	Read bills, invoices, ledger
Financial	<code>financial:edit</code>	Create / modify bills, invoices, ledger entries
Workforce	<code>workforce:edit</code>	Manage workers, attendance, payroll
Supply Chain	<code>supply_chain:edit</code>	Create / edit indents, RFQs, POs, GRNs, stock
Admin	<code>roles:manage</code>	Create / edit / delete custom roles in the org

2.4 Access gates (in evaluation order)

1. **Authentication** — Session cookie + password hash (bcrypt, 12 rounds).
2. **Organisation scope** — Non-super users see only their own organisation's data.
3. **Module access** — Each organisation has an enabled-modules list; each project may override it. Disabled modules return 403.
4. **Project access** — For roles that do not bypass, the user must hold a row in `project_access` for the project.
5. **Capability check** — Specific actions require a capability (built-in preset or granted via custom role).

2.5 Demo accounts

A seed script (`seed-rbac.ts`) provisions one user per built-in role in the *Demo Construction Co* organisation, plus a super-admin in *Mystics Civil HQ*. Shared password: `MysticsCivil@2026` .

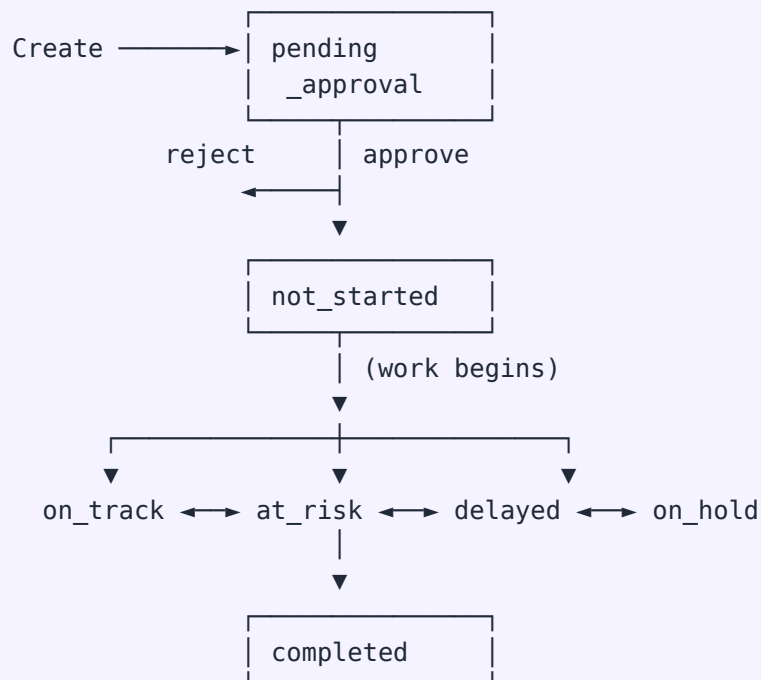
3. Organisation Management

Feature	Who	What
Create / edit organisation	super_admin	Name, legal name, GSTIN, PAN, address
Set project quota (<code>max_projects</code>)	super_admin	Caps how many projects the org may own
Toggle enabled modules	super_admin	Org-wide module visibility
View org stats	admin, super_admin	User count, project count vs quota

Quota enforcement: When `project_count >= max_projects` , new project creation is blocked until the quota is raised.

4. Project Lifecycle

4.1 States



4.2 Workflow

1. **Create** — Anyone with `project:create` files a project. If the creator is `owner` / `admin` / `super_admin`, it is auto-approved into `not_started`. Otherwise it lands in `pending_approval` and an entry is written to the **Approvals Inbox**.
2. **Approve / reject** — A user with `project:approve` (or admin/owner) resolves the inbox item. Approval moves the project to `not_started`; rejection terminates it.
3. **Health transitions** — A user with `project:transition` moves the project between `on_track`, `at_risk`, `delayed`, `on_hold` as conditions change. Each transition is logged.
4. **Resubmit from on-hold** — Bringing a project off hold re-opens its approval workflow if the on-hold was tied to a governance issue.
5. **Complete** — `project:complete` closes the project. Completed projects are read-only.

4.3 Quotas & access

- Each new project counts toward the organisation's `max_projects` quota.
- On creation, the creator is automatically granted project access.
- Admins / owners always see every project in their organisation.

5. Work Breakdown & Schedule

5.1 WBS (`wbs.ts`)

- Hierarchical activities (parent → children, unlimited depth).
- Each activity carries `planned_quantity`, `actual_quantity`, unit, and a cost weight used for earned-value calculations.
- `actual_quantity` is updated automatically when DPRs against that activity are approved.

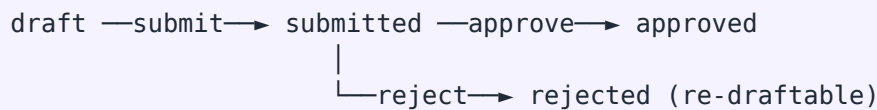
5.2 Milestones (`milestones.ts`)

Status	Meaning
pending	Not yet started
on_track	Progressing per plan
at_risk	Slipping
delayed	Past planned date
completed	Done

Milestones can be linked to WBS activities for roll-up reporting.

6. Daily Progress Reports (DPRs)

6.1 States



6.2 Workflow

1. **Draft** — A user with `dpr:create` (typically site engineer, PM, contractor) creates a DPR for a project + date, logging manpower, weather, quantities executed, narrative remarks.
2. **Attach** — Photos and documents may be attached.
3. **Submit** — User submits; a row is written to the Approvals Inbox.
4. **Approve / reject** — User with `dpr:approve` (typically PM) resolves the inbox item. On **approve**, executed quantities are posted against the corresponding WBS activities (incrementing `actual_quantity`) in a single transaction.
5. **Edit window** — Only `draft` DPRs may be edited (`dpr:edit`).

7. Photos, Documents, Issues

Module	Purpose	Key fields
Photos (<code>photos.ts</code>)	Site documentation, geotagged uploads	project, WBS link, caption, tags, EXIF
Documents (<code>documents.ts</code>)	Drawings, contracts, certificates	versioning, MIME, uploaded-by
Issues (<code>issues.ts</code>)	Site issues / snag list	severity (<code>low</code> → <code>critical</code>), status (<code>open</code> / <code>in_progress</code> / <code>resolved</code>)

All three use **Object Storage** behind the scenes; the API issues short-lived signed URLs (`storage.ts`) for upload and download.

8. Dashboards & Reports

8.1 Dashboard (`dashboard.ts`)

Per-project and org-wide KPIs:

- Active project count vs total
- Pending approvals (across domains)
- DPRs filed today / this week
- Budget vs spend (live from financial module)
- Schedule variance (live from WBS / milestones)

8.2 Reports (`reports.ts`)

Exportable analytics:

- **BOQ vs Actual** — variance by cost head, drill-down to activity.
- **Financial** — bill aging, payment status, GST/TDS summaries.
- **Workforce** — attendance summary, payroll runs.
- **Project health** — milestones at-risk, NCRs open.

Reports render as paginated tables in the UI and can be exported to CSV / PDF.

9. Estimation, BOQ, Rate Engine

9.1 Estimates (`estimation.ts`)

Five-level estimating ladder, each level progressively more detailed:

Level	Use
L0	Order-of-magnitude (square-foot rate)
L1	Parametric, by cost head
L2	Schematic BOQ
L3	Detailed BOQ with rate analysis
L4	Tender BOQ
L5	Contract / awarded BOQ

Each estimate links to **cost heads**, **BOQ items**, and **rate analysis components** (material + labour + plant + overhead build-ups).

9.2 Rate sources (`rate-sources.ts`)

Registered sources of unit rates — DSR (Delhi Schedule of Rates), SSR (State Schedule of Rates), in-house catalogues, vendor quotes. Each source has a type (`csv` , `json` , `gsheet` , `escalation_rule`) and an enable flag.

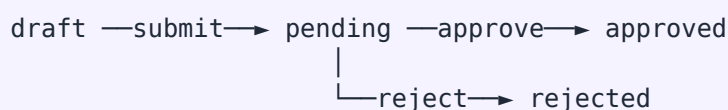
9.3 DSR / SSR rates (`dsr-rates.ts`)

The pulled rate registry. A **daily scheduler** runs at **02:00 server time** (`cron` : `0 2 * * *`) and pulls every enabled source, refreshing rates and logging the import.

Manual sync is also available from the DSR Rates page.

10. Variation Orders (VOs)

10.1 States



`approved` and `rejected` are terminal; the VO cannot be re-opened.

10.2 Workflow

1. **Draft** — QS / PM drafts a VO linked to one or more BOQ items, capturing the scope change, quantity delta, rate, and justification.
2. **Submit** — Status moves to `pending` ; an Approvals Inbox entry is created.
3. **Approve / reject** — A user with `vo:approve` (typically owner, PM, finance) resolves it.
4. **Post-approval** — Approved VOs are reflected in BOQ-vs-Actual reports and downstream contractor billing.

11. Financial Module

The financial module covers contractor billing, payment, ledger, client invoicing and statutory deductions.

11.1 Entities

Entity	Purpose
<code>contractor_bills</code>	Bills raised by contractors / subcontractors
<code>bill_deductions</code>	TDS, retention, advance recovery, LWF, other
<code>payment_vouchers</code>	Payment instruments against approved bills
<code>ledger_accounts</code> , <code>ledger_entries</code>	Double-entry general ledger
<code>client_invoices</code>	Invoices raised to the project's client
<code>gst_entries</code>	Output / input GST register
<code>tds_entries</code>	TDS deducted (194C, etc.)
<code>retention_ledger</code>	Retention held, released schedule
<code>advance_ledger</code>	Mobilisation / material advances and recoveries

11.2 Contractor bill workflow

1. **Draft** — `financial:edit` user creates a draft bill against a project / contractor with line items.
2. **Submit / verify** — Bill moves through verification (QS scrutiny, PM certification) toward approval.

3. **Auto-deductions** — On approval, the system applies:
 - **TDS** (Section 194C, configurable rate)
 - **Retention** (typically 5%, configurable)
 - **Labour Welfare Fund** (where applicable)
 - **Advance recovery** (capped at outstanding advance balance)
4. **GST handling** — Output GST is registered against the bill.
5. **Payment voucher** — Approved bills can be paid via a payment voucher (cash / bank / cheque / NEFT / RTGS).
6. **Ledger posting** — On approval, a double-entry pair is posted: `Dr Expenditure / Cr Contractor Payable` ; on payment, `Dr Payable / Cr Bank` .

11.3 Client invoicing

Symmetric flow for invoices issued to clients, with status tracking (`draft` , `sent` , `paid` , `cancelled`).

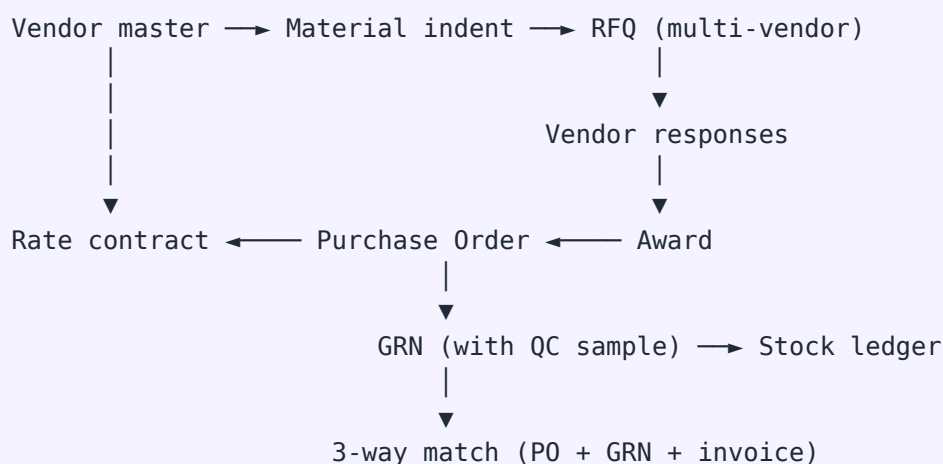
11.4 Visibility

- `financial:view` — read access to bills, invoices, ledger, reports.
- `financial:edit` — create / modify.
- Admins and finance role always have both.

12. Supply Chain Module

End-to-end procurement and inventory.

12.1 Procurement flow



12.2 Entities

Entity	Purpose
vendors	Vendor master with statuses (<code>active</code> , <code>blacklisted</code> , <code>pending_kyc</code>)
material_indents	Site requests for material
rfqs , rfq_responses	Multi-vendor quote collection
purchase_orders , po_items	Awarded POs
grns , grn_items	Goods received, with QC checkpoints
material_tests , test_results	Lab results gating GRN acceptance
stores , inventory_items	Storage locations and stock master
stock_ledger , stock_issues , wastage_log	Inventory movements
rate_contracts	Pre-negotiated rate frames

12.3 Workflow rules

- Indents must be approved before they can be converted to RFQs (above org threshold).
- RFQ awards generate a PO; PO status (`draft` → `issued` → `partially_received` → `closed`).
- GRN acceptance requires QC pass; failed tests raise an NCR (see Quality).
- Each GRN increments stock; each Issue decrements; the **stock ledger** is the immutable audit trail.

13. Workforce Module

13.1 Worker master

14. Quality & Safety

Implemented within the workforce/quality routes (`workforce.ts`).

14.1 Quality

Entity	Purpose
<code>itps</code> (Inspection & Test Plans)	Per-activity checkpoint list; approved before execution starts
<code>itp_items</code>	Individual checkpoints
<code>inspection_requests</code>	Site → QC: "please inspect"
<code>ncrs</code> (Non-Conformance Reports)	Raised on inspection failure
<code>quality_tests</code>	IS-code tests (cube strength, slump, etc.)

NCR lifecycle: `open` → `capa_submitted` → `closed` . CAPA = Corrective And Preventive Action.

14.2 Safety

Entity	Purpose
<code>hira_entries</code>	Hazard Identification & Risk Assessment register
<code>jsa_entries</code>	Job Safety Analysis per activity
<code>safety_permits</code>	Hot work, height work, confined space, electrical — <code>pending</code> → <code>approved</code> → <code>expired</code>
<code>ppe_issues</code>	PPE issued to workers (helmet, vest, gloves, etc.)
<code>incidents</code>	Near-miss / first-aid / lost-time / fatality, with classification

15. Approvals Inbox

A **single inbox** unifies every cross-domain approval task. Domains feeding the inbox include:

- Project creation (`project:approve`)

- DPR submission (`dpr:approve`)
- Variation orders (`vo:approve`)
- Contractor bills (financial approvers)
- Labour contractor bills
- ITPs

Each row carries: source domain, entity id, project, requested-by, requested-at, status (`pending` / `approved` / `rejected`), and a deep link into the source record. Resolving an inbox item performs the underlying domain transition in the same transaction.

16. Modules Access (per-org / per-project gating)

`modules-access.ts` lets administrators turn entire feature areas on or off per organisation and per project.

- **Org default** — Each organisation has an `enabled_modules` list.
- **Project override** — A project may override the org default to enable or disable specific modules (e.g. a small project may turn off the Supply Chain module).
- **Enforcement** — Any API route in a disabled module returns 403; the UI hides the corresponding navigation.

Module keys correspond to the major navigation areas (Projects, DPRs, Financial, Supply Chain, Workforce, Quality, Safety, Estimation, Reports).

17. Admin & Custom Roles UI

The `/admin` page is the control room for administrators.

17.1 Tabs

Tab	Who sees it	Purpose
Users	admin, super_admin	Create / edit / delete users; assign built-in role, organisation, custom role; reset password; manage per-user project access
Custom roles	admin, super_admin, or any user with <code>roles:manage</code>	Create / edit / delete org-scoped custom roles; tick capabilities grouped by domain
Organisations	super_admin only	Set project quota, edit org details

17.2 Custom role dialog

- **Name** — required, unique within org.
- **Description** — optional, ≤ 256 chars.
- **Capabilities** — checklist grouped by domain (Projects, Approvals, DPRs, Financial, Workforce, Supply Chain, Admin).
- **Organisation** — locked on edit; selectable only at creation by super-admin.

Deleting a custom role first detaches every user holding it (their `custom_role_id` is nulled) and then removes the role.

18. Object Storage

`storage.ts` issues short-lived signed URLs to the App Storage bucket. Photos, documents, GRN attachments, NCR evidence, incident photos, and wage slips all go through it. Files are written to either a public path (`PUBLIC_OBJECT_SEARCH_PATHS`) or the private directory (`PRIVATE_OBJECT_DIR`) based on the entity type.

19. Scheduled Jobs

Job	Schedule	Source
Rate sync	Daily at 02:00 server time	<code>runAllEnabledSources()</code> iterates every enabled rate source (DSR, SSR, in-house) and refreshes rates. Failures are logged but do not abort the cycle.

20. Frontend Page Map

Path	Page	Description
/login	login.tsx	Email + password sign-in
/profile	profile.tsx	View own profile, change password
/ (dashboard)	dashboard.tsx	Executive KPIs and pending-approval counts
/projects	projects.tsx	Project list, filters, status badges
/projects/new	project-new.tsx	Project creation wizard
/projects/:id	project-detail.tsx	Tabbed: WBS, milestones, DPRs, settings, modules
/dprs/:id	dpr-detail.tsx	DPR view / edit / submit / approve
/approvals	approvals.tsx	Cross-domain approvals inbox
/variation-orders	variation-orders.tsx	VO list, submit / approve
/estimation	estimation.tsx	L0-L5 estimates, BOQ builder, rate analysis
/dsr-rates	dsr-rates.tsx	DSR / SSR registry, sync controls
/financial	financial.tsx	Bills, vouchers, ledger, invoices
/supply-chain	supply-chain.tsx	Vendors, indents, RFQs, POs, GRNs, stock
/workforce	workforce.tsx	Workers, attendance, payroll, contractor bills, quality, safety
/reports	reports.tsx	Exportable analytics
/boq-vs-actual	boq-vs-actual.tsx	Cost-head variance drill-down
/organisations	organisations.tsx	Organisation directory (super-admin)
/admin	admin.tsx	Users, custom roles, organisations
*	not-found.tsx	404

21. API Route Map

All routes are mounted under `/api`. The proxy middleware (`requireProjectAccess`) automatically gates every `/projects/:projectId/...` route before it reaches the handler.

Route file	Purpose
health.ts	Liveness probe
auth.ts	Login, logout, session
me.ts	Current-user profile, modules, capabilities
admin.ts	User CRUD, organisation CRUD, quota, project access management, assignable-roles catalog
custom-roles.ts	Custom role CRUD + capability catalog
modules-access.ts	Org / project module toggles
organisations.ts	Organisation read endpoints
projects.ts	Project CRUD and lifecycle transitions
wbs.ts	Work breakdown structure CRUD
milestones.ts	Project milestones
dprs.ts	DPR draft / submit / approve / reject
photos.ts	Site photo uploads
documents.ts	Document library + versioning
issues.ts	Site issue tracker
dashboard.ts	KPIs and roll-ups
approvals.ts	Cross-domain inbox
estimation.ts	Estimates, BOQ, rate analysis
dssr-rates.ts	DSR / SSR rate registry
rate-sources.ts	Rate source registry (enable / disable / sync)
variation-orders.ts	VO lifecycle
financial.ts	Bills, deductions, vouchers, ledger, invoices
supply-chain.ts	Vendors, indents, RFQs, POs, GRNs, stock
workforce.ts	Workers, attendance, payroll, contractor bills, ITP, NCR, HIRA, JSA, permits, PPE, incidents, quality tests
reports.ts	Exportable reports

<code>storage.ts</code>	Signed-URL upload / download
<code>geocode.ts</code>	Haversine distance utilities (attendance geofence)

22. End-to-End Project Walkthrough

A typical project from inception to closure:

1. **Setup** (admin / super-admin)
 - Create organisation, set quota, enable modules.
 - Create users; assign built-in roles; optionally a custom role.
2. **Initiate** (PM / owner)
 - Create project → `pending_approval` (or auto-approved if owner/admin).
 - Define WBS and milestones.
 - Build L0-L3 estimate; freeze BOQ at L5.
3. **Procure** (store / PM)
 - Raise material indents, issue RFQs, award POs, receive against GRN with QC checks; update stock ledger.
4. **Execute** (site engineer / contractor)
 - Mark attendance daily (with geofence).
 - File DPRs against WBS activities — quantities feed back into earned value once approved.
 - Photograph progress; raise issues; log incidents and safety permits.
5. **Quality** (QC)
 - Approve ITPs; raise inspection requests; record IS-code tests; open NCRs; track CAPA to closure.
6. **Variations** (QS / PM)
 - Draft VOs for scope changes; route through approval.
7. **Bill & pay** (finance / QS)
 - Contractor bills with auto-deductions (TDS, retention, LWF, advance recovery); approve and pay via voucher; ledger posts automatically.
 - Raise client invoices on milestone completion.
8. **Workforce close-out** (HR / finance)
 - Compute payroll period; approve; release wages.
9. **Health tracking** (PM / owner)

- Transition project among on-track / at-risk / delayed / on-hold as needed; resolve approvals as they queue in the inbox.

10. **Close** (PM / owner)

- Resolve all pending approvals, NCRs, retentions.
- Mark project **completed** . Data becomes read-only.

End of document.